

Akanksha Sachan

CONTACT

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Aging Institute & Dept. of Immunology, School of Medicine, UPitt [LinkedIn](#)
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EDUCATION

University of Pittsburgh '24 – Present
PhD Candidate in Computational Biology
Advised by Dr. Aditi Gurkar & Dr. Jishnu Das

Carnegie Mellon University '22 – '24
PhD Student in Computational Biology
Comp-Bio Department, School of Computer Science | GPA: 3.58/4

Indian Institute of Technology (IIT), Bombay '18 – '22
BTech. in Chemical Engineering | GPA: 8.06/10

AWARDS AND HONORS

American Federation for Aging Research (AFAR) Graduate Scholarship '25

Best Poster Award at the Aging Institute Research Day, UPitt '24

1st Place, Student A100 track NeurIPS LLM Efficiency Challenge | team of 5 '23

\$500 Track Prize (Healthcare Accessibility for Underrepresented Communities), Pitt Challenge Hackathon (6th annual) | team of 3 '22

Recipient of **Undergraduate Research Award** (URA-01) by IIT Bombay '22

Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay for two semesters for financial hardship '21

Awarded a full scholarship (\$5,000) for iSURE undergraduate research program by the University of Notre Dame '20

Top 0.2% (Rank 3188 out of **1.1M+** candidates) in JEE (Main) '18

Top 1.5% (Rank 2517 out of **160k+** candidates) in JEE (Advanced) '18

INVITED TALKS

"FIREFate: Functional and Interpretable Regulatory Encodings of cell Fate" | *paper-in-prep*

International Society for Computational Biology (ISCB)/(ISMB) Jul '25, '26

American Association of Immunologists (AAI) Annual Meeting 2026 Apr '26

Center of Lung Aging and Regeneration, School of Medicine, UPitt Nov '25

30th Annual School of Medicine Graduate Student Research Symposium, UPitt Oct '25

Keystone Symposia: AI in Molecular Biology Sep '25

Joint CMU-UPitt Ph.D. Program in Computational Biology, Orientation Retreat Aug '25

"Metabolic rewiring of skeletal muscle driven by loss of DNA-repair" | *paper-in-prep*

Center for Systems Immunology Annual Retreat, UPitt Nov '25

23rd Annual Department of Immunology Scientific Retreat, UPitt Sep '25

Aging Institute Research Day, UPitt Nov '24

"Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials" | *Publication*

50th European Calcified Tissue Society (ECTS) Congress Apr '23

ATHLETIC ACCOLADES

Bronze Medal | 35th Intercollegiate (inter-IIT) Aquatics Meet | 3rd position in Medley-Relay '19

Swam 12-hours | Swimathon organized by the IIT Bombay aquatics committee '19

Ran 26.2 miles | Dick's Sporting Goods Pittsburgh Marathon | 5 hours 15 mins '24

Ran 13.1 miles | Dick's Sporting Goods Pittsburgh Half Marathon | 2 hours 30 mins '23

3rd Position | Computational Biology Department, Pretty Good Race (5km) for SCS, CMU '23

LEADERSHIP & SOCIETAL CONTRIBUTIONS

Vice President | *Allegheny Science Policy and Governance (ASPG)* Aug '25 – Present

- Leading a graduate-student organization (recognized chapter of the **National Science Policy Network**) bridging the advocacy gap between scientists and policy-makers
- Organized panel discussions with Representatives & Day on the State Capital Hill, Harrisburg, PA for interactions with Senators and discussions on policies for **AI/Data Centers and Aging**

Teaching Assistant | 02-760: *Lab Methods for Computational Biologists*, Fall '23 – Spring '24
SCS, CMU

- Taught a lecture on **genomics technologies** and gene regulation to a class of around 20 students
- Revamped a module of the course based on **RNA-sequencing** analysis of mouse cell lines

Coordinator | *Technights, Carnegie Mellon University, Robotics Institute* Aug '23 – Aug '25

- Led a **computer-science outreach** program teaching middle-school girls technical concepts
- Developed and taught sessions on data analysis, foundation models, and deep learning

Team Lead | *4D Nucleome Consortium Hackathon, Seattle, Washington* Mar '24

- Led a team of 5 during the hackathon to create integrative pipelines for **single-cell HiC** data analysis using deep-learning tools and publicly available cell-atlas scale datasets
- Delivered a 30-minute presentation to showcase results to consortium researchers

Treasurer • Senator | *Graduate Student Association, (CPCB GSA)* Aug '23 – Aug '25

- Managed a budget of **\$5,121** and tracked spending spanning both CMU and UPitt GSA events
- Organized the 2024 open-house event for visiting prospective students admitted to CPCB

Editorial Board Member • Design Convenor | *Insight, the student newsletter of IIT Bombay* '19 – '21

- Sketched & Illustrated **1 Flagship & 1 Special Edition Magazine Cover**, and a merch t-shirt
- Editorial Board member ('20-'21): contributed to editorial oversight and article curation

Convenor, Girls Swim Team | *Aquatics, Indian Institute of Technology Bombay* '19 – '20

- Led logistics, training coordination, and community building events for the institute girls' team

CONSORTIUMS & WORKSHOPS

- SenNet Consortium Annual Meeting** | *Cellular Senescence Network, NIH, Washington, DC* Sep '25
- Attended talks on cellular senescence atlasing and organ-specific senescence signatures
- Grace Hopper Celebration** | *AnitaB.org, Philadelphia* Oct '24
- Experienced talks and brainstorming sessions with tech industry leaders
- Computational Genomics Summer Institute** | *University of California, Los Angeles* Jul '23
- Delivered Lightning Talk: "Connecting 3D genomic organization to gene expression"
- Grad Cohort for Women** | *Computing Research Association (CRA-WP), San Francisco* April '23
- Participated in mentoring sessions from women in CS across industry and academia

RESEARCH EXPERIENCE PRIOR TO THESIS

- Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture** '22 – '24
Supervisor – Prof. Jian Ma, Computational Biology Department, School of Computer Science, CMU
- Developed a framework to identify cis-regulatory, 3-D genomic hubs, controlling target-gene expression and key transcription factors in healthy and cancerous cell lines
 - Implemented **hypergraph-based representation learning** models for integrative analysis of 3D genome architecture in single-cell, multi-omic, mouse brain datasets
- Translational Mechanobiology Lab**, MBI, National University of Singapore '22
Summer Internship | Supervisor – Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine
- Implemented a **Generative-Adversarial-Network (GAN)** pipeline for colorectal cancer imaging data to support robust deep-learning deployment in clinical environments
 - Reviewed domain adaptation and distribution-shift methods for liver fibrosis staging from annotated medical images
- Lab for Bone Biomechanics**, Institute of Biomechanics, ETH Zurich '21
Summer Internship | Supervisor – Prof. Ralph Muller, Dept. of Health Sci&Tech | Presentation
- Added a **Type-2 Diabetes** pathway to an agent-based **osteoporosis** model and ran large-scale simulations on the **CSCS Swiss National Supercomputing Center Cluster**
 - Validated the model against in-vivo data from 235 diabetic patients and identified osteoid production as a key parameter via **sensitivity analysis**
- Multicellular Systems Engineering Lab**, University of Notre Dame '20
Summer Internship | Supervisor – Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering | Poster
- Built a morphometric feature-extraction pipeline for *Drosophila* wing imaginal discs using OpenCV and Fiji to understand organism-wide signaling patterns
 - Evaluated variance under gene-knockout in the Ca^{2+} signaling pathway using PCA-features
- Machine Learning & Statistics**: Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)
- Predicting genomic-hub mediated TF-perturbation response via geometric deep learning** '23
10-708: Probabilistic Graphical Models | Code Repo

KEY COURSEWORK

- Augmented the GEARS GNN for Perturb-seq response prediction with a Hi-C contact graph (K562 **cancer cell-line** Micro-C bulk-seq dataset, 20kb (base-pair) bins mapped to gene TSSs), adding 3D-genome topology to the existing co-expression and Gene Ontology graph features

Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs '23

02-710: *Computational Genomics* | *Code Repo*

- Developed a **GNN** to predict methylation from HiC data, connecting structure-to-phenotype
- Performed clustering with K-means and t-SNE on **single-nucleus DNA methylation** data

Efficient Online Continual Semi-Supervised Learning

'22

10-701: *Intro to ML*

- Improved semi-supervised OCL-PDS with active learning and optimized pseudo-labeling
- In the Amazon-WPDS dataset, our approach outperformed the ER-FIFO method